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Training and Solutions

MuleSoft®

Developer Training

CURRICULUM



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Two-Month Weekend Instructor Lesson Plan

- Beginner to Intermediate
- Job-Oriented Training
- Saturday & Sunday

Duration	7 weeks / 2 months
Final Outcome	Build, test, document and explain a Mule 4 integration project

Course purpose

This instructor-facing plan provides a practical sequence for teaching MuleSoft over seven weekends. It covers Mule 4 fundamentals, flow control, DataWeave, RAML/APIkit, connectors, database and HTTP integration, error handling, MUnit, API security, deployment, monitoring and interview readiness.

WEEK 1

Integration fundamentals and Mule 4 setup

Milestone: Every student runs a local Mule application and tests an endpoint with Postman.

Saturday — Session 1: Integration and MuleSoft fundamentals

- Explain why enterprises integrate systems
- Define APIs, REST, HTTP methods, status codes, headers, request/response, JSON and XML.
- Introduce Anypoint Platform, Mule runtime and API-led connectivity.
- Explain System APIs, Process APIs and Experience APIs using a single business scenario.

Sunday — Session 2: Studio setup and first Mule application

- Set up Anypoint Studio and a compatible Java installation.
- Explain project structure, configuration XML, connectors and Mule runtime.
- Introduce Mule event concepts: payload, attributes and variables.
- Explain sources, processors, flows and subflows.

WEEK 2

Core components and flow control

Milestone: Students can trace a request through a multi-step Mule flow.

Saturday — Session 3 RAML, REST design and APIkit

- Explain REST resource and method design.
- RAML structure, data types, examples and request/response definitions.
- Cover query parameters, URI parameters and error responses.
- Demonstrate APIkit scaffolding and routing from an API specification.

Sunday — Session 4: Mule event and core components

- Explain flow versus subflow and Flow Reference.
- Set Payload, Set Variable and Logger.
- Compare payload, attributes and variables.
- Cover HTTP headers, query parameters, URI parameters and Choice routing.
- For Each, Scatter-Gather concepts and async processing.
- Use Flow Reference to reuse logic.
- Introduce validation and conditional processing.
- Practice debugging and reading application logs.

WEEK 3

DataWeave fundamentals

Milestone: Students can write basic DataWeave mappings independently.

Saturday — Session 5: DataWeave syntax and selectors

- DataWeave headers, output directives, types and expressions.
- Cover objects, arrays, and operators.
- Explain variables, conditions, default values and null handling.
- Discuss common transformation errors and how to read them.

Sunday — Session 6: Mapping functions and format conversion

- Map, filter, mapObject, reduce and distinctBy.
- Cover string, date and numeric functions.
- Demonstrate JSON-to-JSON, JSON-to-XML
- Explain nested-array mapping and output formatting.

WEEK 4

Database and external system integration

Milestone: Students build a Process API that combines data from more than one source.

Saturday — Session 7: Database Connector and SQL

- Review relational database concepts and connection configuration.
- Select, Insert, Update and Delete operations.
- Explain parameterized SQL queries and mapping result sets with DataWeave.
- Discuss database connectivity errors.

Sunday — Session 8: HTTP integration and orchestration

- HTTP Request connector for outbound service calls.
- Configure base URLs, headers, query parameters and timeouts.
- Explain data enrichment, pagination and upstream failures.
- Introduce Salesforce, File and SFTP connector use cases.

WEEK 5

Error handling, security and MUnit

Milestone: Students deliver controlled error responses and an automated test suite.

Saturday — Session 9: Error handling and resilience

- Explain Mule error types and error propagation.
- Compare On Error Continue and On Error Propagate.
- Try scope and global error handlers.
- Cover standard error schemas, logging, timeouts, retries and reconnection.

Sunday — Session 10: Security and automated testing

- Differentiate authentication and authorization.
- Introduce basic authentication concepts, client applications and API policies.
- MUnit structure, assertions, Mock When, Verify Call and negative tests.

WEEK 6

Deployment, API management and troubleshooting

Milestone: Students understand the release workflow and can investigate a common failure.

Saturday — Session 11: Packaging and deployment

- Explain Maven and project dependencies.
- Package a Mule application.
- Introduce Runtime Manager and deployment lifecycle.
- Cover environment properties, secure configuration and CloudHub concepts.

Sunday — Session 12: API Manager, monitoring and job workflows

- Introduce API Manager, API instances, policies and client applications.
- Application logs, runtime status and troubleshooting.
- Introduce CI/CD, Git, pull requests and code review.

WEEK 7

Capstone project and interview readiness

Milestone: Students demonstrate a working, tested and documented integration project.

Session 13: Practical examination and interview preparation

- Review common MuleSoft developer interview concepts.
- Assess DataWeave scenarios, error handling and troubleshooting.
- Conduct project demonstrations and code reviews.
- Give individual feedback and a follow-up learning plan.